

# Malek Swilam

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## EDUCATION

### George Mason University, Honors College

B.S. Computer Engineering

Fairfax, VA

Aug. 2026 – May 2030 (Expected)

### Oakton High School

Advanced Studies Diploma | SAT: 1510 (Math 780, EBRW 730)

Vienna, VA

Aug. 2022 – Jun. 2026

- **Relevant Coursework:** Advanced Data Structures & Algorithms, AP Computer Science A, AP Physics C: Electricity & Magnetism, AP Calculus BC, AP Physics C: Mechanics (exam only), AP Statistics, AP Physics 1

## EXPERIENCE

### Machine Learning Intern

Yanez.ai

Jun. 2026 – Present

Remote

- Built a computer vision ensemble for face anti-spoofing, combining a gradient-boosted model on **50+** engineered image cues with a CNN-based depth model, cutting error **75%** versus the deployed rule-based system on unseen data
- Designed a rank-based score fusion for models with incomparable output scales, adding **3.4 AUC points** over the stronger component and catching **100%** of real attacks that had bypassed the production system
- Boosted throughput **500%+** on an adversarial document tampering pipeline with multi-threading and a load-aware scheduler, and cut cost per image **85–93%** by replacing general-purpose generative models with a dedicated face-swap model

### Engineering Intern

Alarm.com

May 2026

Tysons Corner, VA

- Co-built a Flask lab portal for the hardware QA team with QR-labeled inventory, equipment checkout, admin roles, and Jira-synced request queues, replacing manual tracking of lab equipment
- Soldered buck converters and embedded boards, provisioned Raspberry Pi units, and integrated locks, sensors, cameras, and thermostats into automated hardware testing

### Machine Learning Research Intern

George Mason University, ASSIP

Jun. – Sep. 2025

Fairfax, VA (Hybrid)

- Built an ML pipeline for short-term rainfall forecasting, training and comparing **LSTM**, **XGBoost**, and **ensemble** models on MAE, RMSE, and  $R^2$ ; published in *GMU Journal of Student-Scientists' Research*
- Outperformed both LSTM and XGBoost on every metric by ensembling them, raising  $R^2$  **16–26%** and landing **96%+** of hourly predictions within  $\pm 1\text{mm}$  of actual

### Programming Lead

FTC Robotics, Team 369 (Oakton)

2023 – 2026

Vienna, VA

- Led all robot software development in Java across two varsity seasons; built an autonomous localization pipeline fusing wheel odometry with IMU, with AprilTag detection for field-relative pose correction and motion profiling, ranking in the **top 4%** of **7,000+ teams internationally** for autonomous scoring
- Designed multi-subsystem PIDF control (turret, flywheel, pose stabilization) and a Limelight-based vision targeting system; reached **State Championship** playoffs both seasons and **captained a playoff alliance** at the international **World Championship**

## PROJECTS

### OCRadar: Oral Cancer Detection App | Python, Azure AI Vision, Flutter, Dart

[ocradar.com](https://ocradar.com)

- As **COO and lead Android/Windows developer**, co-built a cross-platform app that screens for early signs of oral cancer using a CNN image classifier on Azure AI Custom Vision with **98% accuracy**; featured by **WTOP News**, **Korea Times**, and Vision Times China

### Voice-Controlled Robotics Agent | Python, MuJoCo, Anthropic API, FastAPI, Whisper [malekswilam.dev/robot-agent](https://malekswilam.dev/robot-agent)

- Built a voice-controlled manipulation agent for a Franka Panda arm in MuJoCo where an LLM decomposes natural-language commands into sub-goals via chain-of-thought reasoning, detects mid-task failures, and replans autonomously
- Solved robot motion via inverse kinematics and streamed the live simulation through a FastAPI WebSocket portal; cut LLM input cost per call **~99%** with prompt caching

### Desk Calendar Agent | Python, Raspberry Pi 5, Local LLM, Google Calendar API

[malekswilam.dev/calendar-agent](https://malekswilam.dev/calendar-agent)

- Built an agent that turns a texted or spoken instruction into a Google Calendar event using a 4-bit quantized 3B-parameter model (Qwen2.5) running **fully on-device** on a Raspberry Pi 5, with deterministic checks that blocked **100%** of destructive actions the model proposed in testing
- Cut on-device prompt processing time **94%** ( $\sim 100\text{s}$  to  $\sim 6\text{s}$ ) with prefix caching; wired the e-ink display to the Raspberry Pi 5, wrote the rendering and display-refresh layer, and selected the hardware components

## TECHNICAL SKILLS

**Languages:** Java, Python, Dart, JavaScript, HTML/CSS

**Technologies:** PyTorch, XGBoost, NumPy, FastAPI, Flask, Flutter, MuJoCo, Azure AI Vision, Anthropic API, Ollama

**Hardware:** Soldering, Embedded Linux, Raspberry Pi, Sensor Fusion & PIDF Control

**Tools:** Git/GitHub, Android Studio, Claude Code, Codex

**Honors:** Eagle Scout; FTC World Championship Alliance Captain; T.R. Cook Award of Excellence in Technology (2025)